Total Error: What Classifier Validation Establishes About Downstream Estimates

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Abstract

Researchers often reuse one predicted item label across many records and construct several aggregates from those records. With skewed item use, one mistake can move several reported estimates together. A confusion matrix records how many mistakes occurred but loses their locations. This note asks what existing validation evidence can establish about estimates built from such labels. For estimates linear in the labels, I give sharp bounds using exact confusion counts and show when those counts identify the estimate. A probability sample of already-verified target items can provide a simultaneous count-based confidence certificate or a direct correction with a joint design covariance for several estimates. In a browsing-data stress test, exact error counts leave the signs of 84 group comparisons unresolved. Its user–domain matrix also shows how record-level independence can misstate prediction-error uncertainty even with independent domain errors. A separate comparison of adult-domain classifiers shows similar information loss with observed disagreements, without treating either classifier as truth. I also distinguish these fixed-label results from score plug-ins, which need relevant conditional mean assumptions, and from regression with aggregated predicted covariates, whose coefficient is nonlinear in the labels. The practical task is to match available validation evidence to the estimates and uncertainty claims the researcher intends to report.

Suppose a researcher classifies reusable items, such as websites, merchants, products, or documents, and links their labels to observational units. The reported quantity might be a total, a difference in means, or a coefficient from regressing an aggregate outcome on observed covariates. The same label may enter many units and several reported statistics, and its influence depends on the particular quantity being estimated. Accuracy and the usual false-positive and false-negative rates count items; they do not record which influential items were misclassified. Knowing the exact confusion matrix for the target collection therefore need not determine even the sign of a group comparison.

The distinction between classifying items well and estimating an aggregate well is established. Work on quantification addresses class prevalence (Hopkins and King, 2010; Forman, 2008; Vaz et al., 2019); cost quantification already allows nonnegative item-specific weights (Forman, 2008). Design-based supervised learning corrects aggregates using probability samples of verified labels (Egami et al., 2023). Election and financial auditing use random checks to control weighted error (Stark, 2008; Shekhar et al., 2023). Machine-generated covariates, including indices assembled from classifications, are studied in regression (Fong and Tyler, 2021; Battaglia et al., 2025; Kluger et al., 2025). Reuse of an uncertain item across records has the same common-shock covariance structure familiar from shift-share designs (Adão et al., 2019). These results motivate a narrower question: for one declared estimate and a fixed set of target items, what does a particular validation summary

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actually establish? Reused labels and signed comparison weights make that question especially consequential.

I answer it for an already assembled dataset, taking its predictions and validation protocol as given. First, I map each linear estimate back to its item weights and derive the smallest interval compatible with exact target-item confusion counts. I characterize when those counts are sufficient. Second, I show how known probabilities for already-verified items give a simultaneous count certificate or a direct correction and joint covariance for declared estimates. Third, I apply the calculations to a browsing-data stress test and to observed disagreements among adult-domain classifiers. The latter is an information exercise because neither classification is adjudicated truth. Aggregated predicted covariates and calibrated scores are related uses of the same labels, but interpreting a regression slope or a score plug-in requires additional assumptions. The note offers a way to match validation evidence to the downstream estimand; it does not introduce a new classifier or a new regression estimator. Choosing which labels to acquire in the first place is the focus of the companion `fewlab` project (Sood, 2026).

1 Reusable labels and downstream targets

Let $y_j \in \{0,1\}$ be the reference label of item j, for $j = 1, \ldots, k$, and let $q_j \in \{0,1\}$ be its prediction. An item can enter many observational units, and a unit can contain many items. A known matrix $C = (c_{ij}) \in \mathbb{R}^{n \times k}$ records how item j contributes to unit i's construct. Its rows might represent people, households, places, or time periods; its columns might represent websites, merchants, products, employers, or documents. Entries can be counts, durations, expenditures, or fixed shares, according to the construct. The complete-label and prediction-based unit vectors are $T = Cy$ and $S = Cq$.

For declared fixed unit weights a, a linear target and its item weights are

$$\theta = a^\mathsf{T} T = w^\mathsf{T} y = \sum_{j=1}^{k} w_j y_j, \qquad w = C^\mathsf{T} a. \tag{1}$$

The weights can be nonnegative for a total or signed for a comparison. Different analyses may use different matrices $C^{(r)}$ and unit weights $a^{(r)}$, while reusing the same predictions q. Each then has item weights $w^{(r)} = C^{(r)\mathsf{T}} a^{(r)}$. Fixed-denominator proportions fit this setup by absorbing the known denominator into the rows of $C^{(r)}$. If a denominator depends on unknown labels, or if the downstream statistic is nonlinear, its target instead has the form $F(C^{(1)}y, \ldots, C^{(R)}y, Z)$ for other observed inputs Z; Section 2 treats a regression example.

A regression of a label-derived outcome on a full-rank, fully observed design X is still linear in the item labels. With $A = (X^\mathsf{T}X)^{-1}X^\mathsf{T}$, its complete-label coefficient is $\beta^* = ACy$. A coefficient contrast $r^\mathsf{T}\beta^*$ has unit weights $a = A^\mathsf{T} r$ and item weights $w = C^\mathsf{T}A^\mathsf{T} r$. These are descriptive coefficients for the specified frame; causal or population interpretations require additional assumptions. Throughout, the link matrices, unit weights, and fully observed regressors used to construct them are fixed. Their uncertainty, if relevant, requires another sampling model.

Browsing is one application. Here c_{ij} can count visits by person i to website j. A difference between group means gives each person in group A weight $1/n_A$ and each in group B weight $-1/n_B$. For browsing shares, divide each row of C by its observed total visits before applying equation (1). Replacing counts by duration changes the exposure unit, and the label definition determines which content the result measures; a domain-level label need not describe every page on a mixed site.

Let $e = q - y$. The prediction-based estimate is $\widehat{\theta} = w^\mathsf{T} q$, so its error is exactly

$$\Delta = \widehat{\theta} - \theta = w^\mathsf{T} e. \tag{2}$$

Thus the estimand determines which classification mistakes matter. A widely used item can dominate a unit's total but cancel from a group comparison if both groups use it equally. A less common item used disproportionately by one group can be influential for that comparison. An error rate that weights all items equally need not describe either target.

The same prediction vector can enter several estimates at once. Let $W = [w^{(1)}, \ldots, w^{(R)}]$ collect the item weights for R declared linear estimates. Their joint error is $W^\mathsf{T} e$, and under a repeated-prediction model its covariance is $W^\mathsf{T} \Sigma W$. The errors in two estimates share the same e_j wherever both weights are nonzero. A common item error may move two group means together and largely disappear from their difference. Appendix D gives their joint covariance under a repeated-prediction model. Retrospective correction from verified items is given in Section 4.

For fixed labels and a fixed fitted classifier, Δ is fixed. If the question instead concerns repeated training and prediction, let $b = \mathrm{E}[q - y]$ and $\Sigma = \mathrm{Cov}(q - y)$ under that process while holding y, w fixed. Then $\mathrm{E}[\Delta] = w^\mathsf{T} b$ and $\mathrm{Var}(\Delta) = w^\mathsf{T} \Sigma w$. Reusing one uncertain item can make errors in different units' aggregates covary even if errors across distinct items are independent. Distinct item errors may also covary, but reuse alone does not imply that they do. These moments require a model of repeated prediction; the count bounds and probability audit below condition on the realized predictions instead. Appendix D gives the covariance details and a rare-error example showing why many items alone do not justify a normal approximation.

2 When an aggregate becomes a regression covariate

Counts and fixed-denominator proportions assembled from predicted labels are also common regression covariates. With fixed aggregation matrix C, their complete-label values are $T = Cy$ and their predicted values are $S = Cq$. Each component of T is linear in item labels. A downstream statistic $F(T, Z)$ need not be: its exact plug-in error is $F(Cq, Z) - F(Cy, Z)$, where Z denotes other fixed inputs. A denominator that depends on predicted labels makes even the underlying proportion nonlinear. The sharp count bounds in Section 3 therefore apply directly only when the declared target is linear in y.

For example, let Y be an observed outcome and V a full-rank matrix of fully observed controls, including an intercept. The coefficient that ordinary least squares would report with complete item labels is

$$\beta(y) = \frac{T^\mathsf{T} M_V Y}{T^\mathsf{T} M_V T} = \frac{y^\mathsf{T} C^\mathsf{T} M_V Y}{y^\mathsf{T} C^\mathsf{T} M_V C y}, \qquad M_V = I - V(V^\mathsf{T} V)^{-1} V^\mathsf{T}, \tag{3}$$

when the denominator is positive. Both numerator and denominator change when labels change, so ordinary attenuation is not a general conclusion. The exact error identity and item-audit moment formulas are in Appendix C. Those formulas do not provide an interval for $\beta(y)$: a ratio of estimated moments can be biased, unstable, or undefined.

This setting intersects work on machine-generated regressors (Fong and Tyler, 2021; Battaglia et al., 2025; Kluger et al., 2025). Battaglia et al. (2025) already study classification and aggregation into regression indices. Kluger et al. (2025) allow nonuniform sampling of complete regression observations; verifying a reusable item here usually supplies only a piece of many observations. A single predicted item label can enter several units' indices. This creates the same overlap structure as a shift-share regressor (Adão et al., 2019): if item errors are random, independent across items, and have variances v_j, then the error in two units' covariates has covariance $\sum_j c_{ij} c_{\ell j} v_j$. Independence of the item errors therefore does not give independence of the regression rows. The overlap can matter in either direction for a coefficient. An item with the same weight in every unit affects their totals, but its error disappears from a regression with an intercept because $M_V C_{\cdot j} = 0$.

There is also a generated-regressor issue beyond the covariance of the rows. The OLS numerator is linear in the labels but its denominator contains $G = C^\mathsf{T} M_V C$, including cross-products of items that enter the same units after residualizing on controls. Appendix C gives the exact plug-in score and its expectation under a specified prediction-error process. If the same predictions build two or more covariates, their measurement errors also enter the off-diagonal cross-products of the regression's normal equations. In that case there need not be a single attenuation factor: an error shared by two covariates can move both coefficients. The relevant cross-products are given in Appendix C. A verified item can inform many units' aggregates, but usually does not reveal any one unit's entire covariate. Methods requiring complete covariates for sampled regression rows therefore need a different validation design or an adaptation to item-level labels. The companion `fewlab` project develops such an item design and analyzes a constructed covariate under its randomized label-acquisition process (Sood, 2026). For equation (3), an item audit must account for item pairs in the quadratic denominator; correcting each aggregate separately is not enough.

A model-based target requires different assumptions. If information $\mathcal{H}$ contains C, V and the prediction inputs, and $p_j = \mathrm{E}[y_j \mid \mathcal{H}]$, then $Cp = \mathrm{E}[T \mid \mathcal{H}]$. Under $Y = V\gamma + \beta T + \varepsilon$ with $\mathrm{E}[\varepsilon \mid \mathcal{H}] = 0$ and a full-rank design, regressing Y on (V, Cp) identifies the population β. Marginal score calibration across items does not establish these conditional means or the exclusion assumption. This population coefficient is also distinct from the fixed-sample complete-label coefficient in equation (3).

3 What a confusion matrix identifies

A confusion matrix can be exactly correct for the target items and still fail to determine a downstream estimate. This is an information problem before it is a sampling problem. Fix the predictions q and the item weights w in equation (1). We now ask what a validation summary establishes about the fixed target $\theta = w^\mathsf{T} y$, without specifying a stochastic model for how the classifier's mistakes were generated.

Let $P = \{j : q_j = 1\}$ and $N = \{j : q_j = 0\}$. Suppose the entire target-population confusion matrix is known: f predictions in P are false positives and g predictions in N are false negatives. This grants more information than ordinary held-out validation, whose error rates require a transport assumption. For a finite collection B of weights, let $S_-(B, m)$ and $S_+(B, m)$ denote the sums of its m smallest and largest elements, with a zero sum when $m = 0$.

Proposition 1 (Sharp bounds from confusion counts)**.** *Across binary label vectors compatible with q, f, g, the smallest and largest possible errors $\Delta = \widehat{\theta} - \theta$ are*

$$\begin{aligned} L &= S_-(w_P, f) - S_+(w_N, g), \\ U &= S_+(w_P, f) - S_-(w_N, g). \end{aligned} \tag{4}$$

Consequently $[\widehat{\theta} - U, \widehat{\theta} - L]$ is the smallest interval containing every compatible value of θ. Both endpoints are attainable.

Proof. Only false positives and false negatives contribute: $\Delta = \sum_{j \in \mathrm{FP}} w_j - \sum_{j \in \mathrm{FN}} w_j$. Select f indices from P and g from N. The two selections have disjoint supports and no other restrictions. Selecting the smallest positive-stratum weights and largest negative-stratum weights minimizes the error; reversing these selections maximizes it. Assign the selected indices the corresponding true labels to attain each endpoint. □

The compatible values form a finite set, so the interval is its convex hull; not every interior value need be attainable. A reported sign is established by these counts only if the bounding interval excludes zero. The calculation requires sorting, rather than a general integer-programming solver.

For example, take $q = (1, 1, 0, 0)$, $w = (1, -1, 1, -1)$, and $f = g = 1$. The same confusion matrix permits errors of -2, 0, and 2. Prevalence, accuracy, FPR, and FNR are identical across those assignments. Adding correctly classified items with zero weight makes accuracy arbitrarily close to one without resolving the target. In the browsing application, these can be domains that affect individual exposure but cancel from a group contrast.

Proposition 2 (When counts suffice)**.** *In a stratum of M items sharing the same predicted label, with exactly m errors, the error sum is point identified if and only if $m \in \{0, M\}$ or all its contrast weights are equal. Its identification width obeys*

$$S_+(w, m) - S_-(w, m) \leq \min(m, M - m)\{\max_j w_j - \min_j w_j\}. \tag{5}$$

Widths add across strata. For several comparisons, counts determine every comparison if and only if their vector of item weights is constant within each stratum with an interior error count.

Proof. Constant weights, no errors, and all errors each determine the sum. If $0 < m < M$ and two weights differ, choose an error set containing one of those indices but not the other. Exchanging them changes the sum, proving necessity. The difference between two sums of m terms is at most m times the range. Taking complements gives the corresponding bound with $M - m$ terms. The vector statement applies the exchange argument to each coordinate. □

For a stratum mixing predicted labels, the corresponding criterion applies to the signed contributions $(2q_j - 1)w_j$, not to w_j alone. The result specifies what a finer validation table must preserve. Stratifying by predicted label alone is sufficient only when the contrast attaches equal weights within each partially erroneous stratum. Partitioning items by similar contrast weights can reduce ambiguity; bins chosen for one comparison need not work for another. In browsing, error rates conditional on user demographics are not automatically available: the same domain can enter many groups.

4 What collected validation labels can support

An existing probability sample of verified target items can recover information that a confusion matrix discards; it need not have been allocated optimally. The construction uses established finite-population counting and audit ideas (Stark, 2008; Shekhar et al., 2023). Partition the target items using information known before labels are audited, then draw a fixed-size random sample without replacement within each stratum. Exact hypergeometric inversion gives a confidence interval $[\ell_h, u_h]$ for the number of incorrect predictions in stratum h; allocate error probabilities across strata so their sum is at most δ.

Let O contain the audited items and $\mathcal{Y}_A$ the binary label vectors that match their verified labels and whose error counts in each stratum lie in $[\ell_h, u_h]$. Its construction and the sorting formula for its extremes are in Appendix E.

Proposition 3 (A simultaneous count certificate)**.** *For any fixed item weights w, the interval $[\min_{z \in \mathcal{Y}_A} w^\mathsf{T} z, \max_{z \in \mathcal{Y}_A} w^\mathsf{T} z]$ covers the fixed target $w^\mathsf{T} y$ with probability at least $1 - \delta$ over the audit design. The same event covers every linear contrast of this one label vector. Its endpoints are sharp given the audited labels and the count intervals.*

The certificate is simultaneous because the count event constrains the same label vector for every w. It can nevertheless be wide: it retains error counts but discards the weights attached

to unaudited error locations. The example and full audit comparison in Appendix E show this loss. Those simulations evaluate a design chosen for one coefficient; they do not establish efficient simultaneous auditing for other contrasts.

For one declared contrast, a stratified difference estimator instead keeps the audited item identities and their weighted residuals:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j \in O_h} w_j (y_j - q_j). \tag{6}$$

With fixed predictions and verified labels, this estimator is design unbiased. A normal interval needs a separate approximation. The conservative finite-sample Hoeffding interval in the appendix is wide in this application; the paper does not establish an efficient finite-sample interval or an optimal audit allocation. The distinction here is between preserving the locations of audited errors and reducing them to counts.

The same collected labels can correct several estimates jointly. Let W contain their item-weight vectors, let R_j indicate that item j was verified, and let $\pi_j > 0$ be its known inclusion probability. Set $g_j = (y_j - q_j) W_{j\cdot}^{\mathsf{T}}$. With predictions frozen before the validation sample was selected,

$$\widetilde{\boldsymbol{\theta}} = W^{\mathsf{T}} q + \sum_j \frac{R_j}{\pi_j} g_j, \qquad \mathrm{E}_R[\widetilde{\boldsymbol{\theta}}] = W^{\mathsf{T}} y. \tag{7}$$

The reusable item, not each downstream record, is the sampling unit. If $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1)$ and $\pi_{jj} = \pi_j$, the exact joint design covariance is

$$\mathrm{Cov}_R(\widetilde{\boldsymbol{\theta}}) = \sum_{j,\ell} \frac{\pi_{j\ell} - \pi_j \pi_\ell}{\pi_j \pi_\ell} g_j g_\ell^{\mathsf{T}}. \tag{8}$$

If every relevant $\pi_{j\ell} > 0$, replacing each summand by $R_j R_\ell / \pi_{j\ell}$ times that summand gives an unbiased covariance estimator. For independent Bernoulli item selection, only the diagonal terms remain, giving $\sum_j (1 - \pi_j)(y_j - q_j)^2 W_{j\cdot}^{\mathsf{T}} W_{j\cdot} / \pi_j$. The off-diagonal entries of this covariance describe how a single verified item corrects several reported estimates together. Highly influential items with small π_j can leave a valid estimator with a very unstable variance estimate and a poor normal approximation.

If validation items were chosen purposively and their inclusion probabilities are unknown, these design claims do not apply. One can instead state a model for how validation errors transfer to the target items, or report bounds under explicit restrictions. A confusion matrix from a convenience set does not identify either the target-item correction or its standard error. The ordinary regression standard error conditional on frozen predictions answers a different question; it does not include the error from unverified labels. If the same selected labels were used to fit the predictions, the frozen-proxy design argument also needs modification; cross-fitting or joint analysis of training and validation would require its own justification.

5 What calibrated scores would add

The bounds from exact target-item confusion counts and the probability audit above condition on fixed predictions and labels. Neither requires calibrated classifier scores. The audit needs a known probability-sampling design for target items and verified labels for the sampled items. If the error counts come instead from a separate validation collection, applying them to the target items requires a transport assumption, whether or not that collection's scores are calibrated.

A score-based plug-in uses a different source of information. Let $p_j \in [0,1]$ be a score intended to represent the probability that target item j has label one, and put $\theta_p = w^\mathsf{T} p$. To interpret this as a conditional mean, introduce a model in which the labels are random given information $\mathcal{I}$ containing the scores and the target weights. Write $\mu_j = \mathrm{E}[y_j \mid \mathcal{I}]$. Then

$$\mathrm{E}[\theta_p - \theta \mid \mathcal{I}] = w^\mathsf{T}(p - \mu). \tag{9}$$

The plug-in equals $\mathrm{E}[\theta \mid \mathcal{I}]$ if and only if this weighted error is zero. A sufficient condition is $p_j = \Pr(y_j = 1 \mid \mathcal{I})$ for each target item. For a particular target, offsetting score errors can also make the weighted sum zero without item-level conditional calibration.

Ordinary calibration, $\Pr(y = 1 \mid p = s) = s$ across items, is weaker. For example, let half the items have label one and weight one, and the other half have label zero and weight zero. Give every item score $1/2$. The scores are empirically calibrated in the unweighted item collection, but θ_p is only half of the realized target. Conditioning on features that distinguish the two weight groups, or directly checking the weighted residual, exposes the failure. Calibration assessed on a separate collection also has to transfer to the target items and their label definition.

For a browsing visit share, $w_j = v_j / \sum_\ell v_\ell$ gives $\theta_p = \sum_j v_j p_j / \sum_j v_j$. Conditional probabilities for the specified content label would make this the model's expected visit share. A group contrast or regression coefficient has different, possibly signed weights; its required weighted score condition changes accordingly.

Even valid target-item conditional probabilities identify a conditional expectation, not the realized finite-item target. Under the conditional-label model, uncertainty around that expectation depends on the joint distribution:

$$\mathrm{Var}(\theta \mid \mathcal{I}) = w^\mathsf{T} \mathrm{Cov}(y \mid \mathcal{I}) w. \tag{10}$$

Item-level calibration supplies neither the off-diagonal covariances nor, by itself, a coverage guarantee for an interval about the realized target. Estimating the scores introduces further uncertainty. These model-based claims are distinct from the fixed-label, design-based coverage of the audit.

6 A browsing-data stress test

The empirical question is how much information confusion counts retain for comparisons supported by an actual browsing matrix. The fixed benchmark contains 1,132 pseudonymized users, 34,078 domains, 103,853 user–domain records, and 4,767,099 visits. It is the bundled `fewlab` extract of the June 2022 YouGov Pulse/RealityMine panel with Blacklight scanner outcomes. The replication package pins the source commit and file hashes. Its analytic universe contains domains with recorded scanner labels; exposure shares use visits within that universe as their denominator. These recorded outcomes are reference labels for the experiment, not error-free measures of all tracking activity. The exercise makes no causal or population-representative claim about demographic differences.

We retain the extract's demographic design matrix: an intercept and indicators for gender, race, education, and source-coded age groups. Omitted categories are Male, White, HS or Below, and under 25. For each of seven scanner outcomes, we calculate all twelve non-intercept least-squares coefficients for both visit totals and visit shares. Each coefficient is a fixed weighted label sum. Two scanner columns are counts and are dichotomized at a positive value; the other five are binary. Complete labels, valid joins, positive visits, and full column rank are checked before estimation.

Predictions are imposed, rather than produced by a fitted classifier. Within each true class, we randomly flip the integer part of 1%, 5%, or 10% of the labels. We then perform two distinct exercises.

First, holding those predictions fixed, Proposition 1 gives the true-coefficient range compatible with their exact confusion counts. Second, holding the reference labels fixed, we rearrange the same numbers of false positives and false negatives to minimize or maximize the predicted coefficient. The latter asks whether two classifiers with identical confusion matrices can report opposing comparisons. It is a worst-case sensitivity analysis, not an estimate of likely classifier bias. Extrema are computed separately for each coefficient; a single error arrangement need not reverse all comparisons at once.

Table 1: What exact confusion counts establish at 1% imposed error in each true class

Scanner outcome	Sign identified	Median realized error (pp)	Median bound width (pp)
Advertising trackers	0/12	0.43	47.8
Third-party cookies	0/12	0.09	52.5
Canvas fingerprinting	0/12	0.34	29.5
Session recording	0/12	0.40	31.6
Key logging	0/12	2.10	21.3
Facebook pixel	0/12	0.10	43.6
Google Analytics	0/12	0.33	21.6

Each row covers twelve demographic coefficients of visit shares in the fixed browsing frame. "Sign identified" counts bounding intervals excluding zero for fixed random predictions. Realized error is the median absolute difference between the prediction-based and reference coefficients. Bound width is the median identification-interval width; both are in percentage points (pp). Error counts are rounded down within each true class. These are not sampling intervals. All outcomes, coefficients, scales, and error fractions are in the machine-readable results.

At the smallest imposed error fraction, none of the 84 share-coefficient signs is identified by the full confusion matrix (Table 1). The realized errors from random placement are usually much smaller than the count-compatible ranges. For example, the reference female coefficient for Facebook-pixel visit share is 0.828 percentage points. The fixed random predictions differ by 0.040 percentage points, while their count-compatible interval for the reference coefficient is $[-17.00, 16.72]$ percentage points. Moving the same numbers of errors can reverse all 84 share-coefficient signs, but those placements are adversarial. These results show what the confusion counts fail to rule out for this frame, not what a fitted classifier is likely to do. Total coefficients and larger imposed error fractions are in the replication results.

How much does domain reuse matter for uncertainty? The incidence matrix permits a separate diagnostic without fitting a classifier. Suppose each domain's prediction error is independent of the others and has the same variance, as under independent binary flips with an equal false-positive and false-negative probability. Equation (D9) compares the prediction-error variance of a linear user statistic with a calculation that keeps each user's marginal error variance but discards covariance between users. For the mean visit share the variance ratio is 323.4, and one domain accounts for 81.2% of the item-level variance under this model. The female and Black share-coefficient ratios are 2.12 and 0.77: shared domain errors can cancel in signed contrasts. These ratios describe the observed weights under stipulated errors, not uncertainty for a fitted classifier. More directly, the same predicted domain labels enter the female and male mean visit shares. Their prediction errors have correlation 0.987 under the equal-variance item model. The error variance of the female-minus-male difference is only 1.36% of a calculation that treats the two group-mean errors as independent. Skew in the domain weights produces strong common movement in the two means, while differencing removes much of it.

An audit can retain more. Appendix E evaluates count projection and direct weighted residual estimation on the same probability-audit draws for one declared coefficient. The count certificate covered that coefficient in every draw but never settled its sign. The direct normal intervals were narrower but undercovered in the smallest random design; the finite-sample Hoeffding baseline was wider than count projection. The exercise illustrates the information lost by counts and the limits of these particular intervals. It does not compare efficient finite-sample procedures or assess the theorem's simultaneous utility across the other coefficients.

7 Observed disagreement between domain classifiers

The imposed-error experiment controls the number and locations of mistakes. A separate comparison uses observed outputs from the adult-domain study of Sood and Shen (2024). Its public domain summary records 6,236,834 visits to 64,074 domains in the June 2022 YouGov panel. The study used a Piedomains rule that marks a domain when its predicted class is "porn" and its reported class probability is at least 0.033. We compare that rule with recorded "porn" categories from Bitdefender, "sex" from Forcepoint, and "Pornography" from alphaMountain. Each row retains only domains with a recorded category from that provider, and visit shares use the visits in that row's frame as their denominator. Missing provider categories are not negative labels.

Table 2: Observed classifier disagreement and visit-weighted shares

Reference	Domains	Visits (%)	Agree (%)	Domain + (%)	Visit + (%)	Count range (%)
Bitdefender	43,391	84.2	97.7	1.92/3.36	1.05/1.71	[0.07, 82.11]
Forcepoint	47,950	87.5	98.0	1.65/3.04	1.02/1.64	[0.07, 80.61]
alphaMountain	36,063	52.7	99.0	0.98/1.35	1.38/2.16	[0.04, 68.07]

Piedomains is the proxy in every row; the named provider supplies the reference category. "Visits" is the share of all 6,236,834 visits represented by domains with a recorded reference category. Agreement counts domains equally. "Domain +" and "Visit +" list Piedomains/provider percentages, first for domains counted equally and then for visits within that row's frame. The count range is the sharp range of the provider visit share if only the two directional disagreement counts are kept, holding Piedomains labels and visit weights fixed. The providers use different category definitions; none is an adjudicated truth label. A recorded category other than the named positive one counts as negative for this comparison, including nonspecific categories such as "misc."

On the 43,391 domains categorized by Bitdefender, which account for 84.2% of visits, the two classifications agree on 97.7% of domains. Piedomains marks 1.92% of domains positive, compared with 3.36% under Bitdefender's category. Their visit-weighted shares are 1.05% and 1.71%, respectively. The discrepancy is visible before weighting domains by visits, so this comparison does not isolate an effect of visit concentration. There are 193 Piedomains-only and 816 Bitdefender-only positive domains.

With both classifications observed, the provider's visit share is known. If the disagreement locations were discarded and only their directional counts retained, its compatible range would be [0.07, 82.11]%. A single Piedomains-negative domain carries 18.8% of visits in this frame and can dominate the count-only upper bound. Appendix F shows how that range changes when the observed label pairs of the most-visited domains are retained. This isolates information discarded by a confusion table; it does not establish that the extreme allocations are plausible classifier errors.

This is a comparison between imperfect classification systems, not an estimate of either system's error against verified content labels. The public repository files used here do not include the person–domain matrix needed to apply the same calculation to the adult study's partisan coefficients. A probability audit of domains in that matrix, with an explicit content-labeling rule, would support a direct validation analysis of those coefficients without assuming calibrated scores. The reported Piedomains class probability and the provider categories alone do not establish the conditional probabilities required for the score plug-in in Section 5.

8 Implications for validation

After data collection, retain the identity of each verified item and how it was selected. For declared linear targets, known item inclusion probabilities support a joint correction and design covariance

even when the validation allocation was inefficient. Exact confusion counts instead constrain the number of mistakes but generally not their weighted sum; their sharp bounds state what those counts alone establish. If several targets matter, their different weight vectors determine which error locations the validation data must preserve. Heavy item influence can make the covariance estimate unstable and a normal interval unreliable even when the design formula is unbiased.

These fixed-label procedures need verified labels and, for an audit, a known sampling design. They do not require calibrated scores. Applying error rates from another collection requires transport assumptions; interpreting scores as conditional probabilities requires stronger conditions described in Section 5. For aggregated predicted covariates, estimating their components does not by itself give valid inference for a regression slope: numerator and denominator moments, their joint uncertainty, and a nonvanishing denominator all matter. The browsing-specific common-rate correction and its assumptions appear in Appendix A.

References

Rodrigo Adão, Michal Kolesár, and Eduardo Morales. Shift-share designs: Theory and inference. *The Quarterly Journal of Economics*, 134(4):1949–2010, 2019. doi: 10.1093/qje/qjz025.

Laura Battaglia, Timothy Christensen, Stephen Hansen, and Szymon Sacher. Inference for regression with variables generated by AI or machine learning, 2025. URL https://szymon.info/assets/pdf/BCHS.pdf. Working paper, April 29 version.

Naoki Egami, Musashi Hinck, Brandon Stewart, and Hanying Wei. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 68589–68601. Curran Associates, Inc., 2023. doi: 10.52202/075280-3000. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/d862f7f5445255090de13b825b880d59-Abstract-Conference.html.

Christian Fong and Matthew Tyler. Machine learning predictions as regression covariates. *Political Analysis*, 29(4):467–484, 2021. doi: 10.1017/pan.2020.38.

George Forman. Quantifying counts and costs via classification. *Data Mining and Knowledge Discovery*, 17 (2):164–206, 2008. doi: 10.1007/s10618-008-0097-y.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

Daniel J. Hopkins and Gary King. A method of automated nonparametric content analysis for social science. *American Journal of Political Science*, 54(1):229–247, 2010. doi: 10.1111/j.1540-5907.2009.00428.x.

Dan M. Kluger, Kerri Lu, Tijana Zrnic, Sherrie Wang, and Stephen Bates. Prediction-powered inference with imputed covariates and nonuniform sampling, 2025. Working paper.

Suriyan Laohaprapanon and Gaurav Sood. Domain knowledge: Predicting the kind of content hosted by a domain, 2020. URL https://github.com/themains/domain_knowledge/blob/850644c06d19a3d787a444443472367dd1bd3046/ms/domain_knowledge.tex. Manuscript version committed June 14. Section: Browsing Data: Concerns and Solutions.

Shubhanshu Shekhar, Ziyu Xu, Zachary Lipton, Pierre Liang, and Aaditya Ramdas. Risk-limiting financial audits via weighted sampling without replacement. In *Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence*, volume 216 of *Proceedings of Machine Learning Research*, pages 1932–1941, 2023. URL https://proceedings.mlr.press/v216/shekhar23a.html.

Gaurav Sood. Estimand-driven sampling of reusable labels, 2026. URL https://github.com/finite-sample/fewlab. Unpublished working manuscript and companion software, version consulted October 4, 2026.

Gaurav Sood and Lucas Shen. Holier than thou? no large partisan gaps in the consumption of pornography online. *Journal of Quantitative Description: Digital Media*, 4, 2024. doi: 10.51685/jqd.2024.011.

Philip B. Stark. Conservative statistical post-election audits. *The Annals of Applied Statistics*, 2(2):550–581, 2008. doi: 10.1214/08-AOAS161.

Afonso Fernandes Vaz, Rafael Izbicki, and Rafael Bassi Stern. Quantification under prior probability shift:

the ratio estimator and its extensions. *Journal of Machine Learning Research*, 20(79):1–33, 2019. URL https://www.jmlr.org/papers/v20/18-456.html.

Appendix

A Bias and the denominators of error rates

Let α_j be the probability of predicting one for a truly negative domain and β_j the probability of predicting zero for a truly positive domain, conditional on the browsing design. Then

$$b_j = \alpha_j(1 - y_j) - \beta_j y_j, \qquad \mathrm{E}[d_i \mid C, y] = \sum_j c_{ij}\{\alpha_j(1 - y_j) - \beta_j y_j\}. \tag{A1}$$

False-positive rates condition on true negatives; false-negative rates condition on true positives. Multiplying a false-positive rate by predicted positives uses the wrong denominator. Such a calculation would instead require the probability of a true negative conditional on a positive prediction, together with assumptions that justify transferring that probability to the weighted population. These conditional probabilities are not interchangeable.

If $\alpha_j = \alpha$ on negative domains and $\beta_j = \beta$ on positive domains,

$$\mathrm{E}[d_i \mid C, y] = \alpha(V_i - T_i) - \beta T_i = V_i\{\alpha - (\alpha + \beta)\pi_i\}, \qquad \pi_i = T_i/V_i, \tag{A2}$$

where the share π_i is defined for $V_i > 0$. At a fixed true share, expected bias scales with browsing volume unless the expression in braces is zero. For $\alpha + \beta > 0$, cancellation occurs at $\pi_i = \alpha/(\alpha + \beta)$. Equal positive error rates cancel only when half of visits are to truly positive domains. Thus the difference between FPR and FNR alone does not determine bias. Nor does the number of distinct visited domains: repeated visits matter through c_{ij}.

The common-rate assumption is stronger than knowing an average error rate on a validation set of domains. Errors may depend on popularity, domain type, or features associated with a particular group's browsing. In that case, equation (A1), with rates conditional on those features, is the relevant expression. The bias in a share is $\mathrm{E}[d_i \mid C, y]/V_i$; dividing by volume removes the mechanical scale factor, not misclassification bias.

With known α, β and $\alpha + \beta \neq 1$, this model gives an unbiased correction:

$$\widetilde{T}_i = \frac{\widehat{T}_i - \alpha V_i}{1 - \alpha - \beta}, \qquad \mathrm{E}[\widetilde{T}_i \mid C, y] = T_i. \tag{A3}$$

The correction is unstable when the denominator is close to zero and may return values outside $[0, V_i]$. Clipping changes its expectation. Estimating error rates introduces additional uncertainty shared across users; rates estimated on unweighted domains need not transfer to weighted exposure. This is the connection to classifier-based quantification (Vaz et al., 2019), with the additional requirement that the rates apply to each user's weights.

B A connection to label sampling

The same mapping supplies a remedy when a researcher can verify a probability sample of items. The companion `fewlab` project develops sampling designs for reusable labels (Sood, 2026). In its browsing application, the regression decomposition is $H = AC$, with column h_j giving the influence of domain j on the coefficient. For exposure shares, replace C by its row-normalized version. For any scalar target in equation (2), the corresponding item influence is w_j.

Freeze predictions q_j before drawing a label sample. Let R_j indicate selection, with known inclusion probability $s_j > 0$ for every item. Conditional on the fixed population and predictions,

the difference estimator is

$$\widetilde{\theta} = \sum_j w_j q_j + \sum_j \frac{R_j}{s_j} w_j (y_j - q_j). \tag{B1}$$

Since $\mathrm{E}_R[R_j/s_j] = 1$, its expectation over label sampling is $\sum_j w_j y_j = \theta$, regardless of the classifier's errors. Verified labels must measure y_j correctly. Retraining the proxy on the same sampled labels requires a separate argument; the fixed-proxy proof does not cover it.

Under independent Bernoulli item sampling,

$$\mathrm{Var}_R(\widetilde{\theta}) = \sum_j \frac{1 - s_j}{s_j} w_j^2 (y_j - q_j)^2. \tag{B2}$$

The realized sample gives an unbiased variance estimator by replacing each summand with $R_j(1 - s_j)w_j^2(y_j - q_j)^2/s_j^2$. Fixed-size or balanced designs require their joint-inclusion terms or a justified approximation. An inaccurate proxy can increase variance relative to omitting it, but does not introduce design bias under these assumptions. This is uncertainty from randomized label acquisition, with y and q fixed; it is distinct from the prediction-error process in equation (D4). Both calculations preserve the reuse of an item label across downstream records.

C Auditing an aggregated regression covariate

The general plug-in error is $F(Cq, Z) - F(Cy, Z)$. If scalar F is differentiable throughout the segment from Cy to Cq, it also has the exact representation

$$F(Cq, Z) - F(Cy, Z) = \int_0^1 \left[C^\mathsf{T} \nabla_T F\{Cy + sC(q - y), Z\} \right]^\mathsf{T} (q - y)\, ds. \tag{C1}$$

Unlike the fixed weights of a linear target, these effective item weights depend on unknown labels along the path. Thresholds need not be differentiable.

For the OLS coefficient in equation (3), set $d = C(q - y)$, $t = M_V C y$, $h = M_V d$, and $u = M_V Y - \beta(y)t$, so $t^\mathsf{T} u = 0$. If the predicted-regressor denominator is positive, the exact error is

$$\beta(q) - \beta(y) = \frac{h^\mathsf{T} u - \beta(y)\{t^\mathsf{T} h + h^\mathsf{T} h\}}{(t + h)^\mathsf{T}(t + h)}. \tag{C2}$$

The quadratic term can remain when first-order terms cancel, and the coefficient can move in either direction.

The conventional OLS score makes the two channels explicit. Write $a = C^\mathsf{T} M_V Y$, $G = C^\mathsf{T} M_V C$, and $U(z; \beta) = z^\mathsf{T} a - \beta z^\mathsf{T} G z$. For the fixed-sample complete-label coefficient $\beta^* = \beta(y)$, $U(y; \beta^*) = 0$, whereas

$$U(q; \beta^*) = e^\mathsf{T}(a - 2\beta^* G y) - \beta^* e^\mathsf{T} G e, \qquad e = q - y. \tag{C3}$$

If repeated predictions are meaningful while y, C, V, Y remain fixed, let $b = \mathrm{E}[e]$ and $\Sigma = \mathrm{Cov}(e)$. Then

$$\mathrm{E}[U(q; \beta^*)] = b^\mathsf{T}(a - 2\beta^* G y) - \beta^*\{b^\mathsf{T} G b + \mathrm{tr}(G\Sigma)\}. \tag{C4}$$

The trace term equals $\mathrm{tr}(M_V C \Sigma C^\mathsf{T})$. With independent item errors it is $\sum_j v_j \|M_V C_{\cdot j}\|^2$: each item's contribution depends on variation in its unit-level loading after removing the controls. It is nonnegative, but the other terms in equation (C4) can have either sign, and an expected score is not the expectation of the OLS coefficient. In particular, ordinary attenuation does not follow. For fixed binary y and hard binary q, nontrivial errors cannot have $b = 0$ item by item: a negative item

can only have a positive error, and a positive item can only have a negative one. Mean-zero error is thus not a harmless default for this prediction process.

For several covariates built from the same predictions, let $T_r = C^{(r)}y$ and $d_r = C^{(r)}e$ on the same regression rows. The (r, s) error cross-product in their residualized normal equations has expectation

$$\mathrm{E}[d_r^{\mathsf{T}} M_V d_s] = b^{\mathsf{T}} C^{(r)\mathsf{T}} M_V C^{(s)} b + \mathrm{tr}\{C^{(r)\mathsf{T}} M_V C^{(s)} \Sigma\}. \tag{C5}$$

Even if distinct item errors are independent, the trace term contains $\sum_j v_j (M_V C^{(r)}_{\cdot j})^{\mathsf{T}} (M_V C^{(s)}_{\cdot j})$. One item prediction can therefore distort the cross-product between two constructed regressors. If the outcome also uses these predictions, its measurement error contributes the analogous term $\mathrm{E}[d_r^{\mathsf{T}} M_V d_0]$ to the regression numerator, with $C^{(0)}$ building the outcome. Shared errors can then create an apparent association as well as distort a regressor's denominator. These terms depend on how each aggregation weights the same item; they need not produce scalar attenuation of any coefficient.

More generally, let $D(y)$ collect p columns of the form $C^{(r)}y$. With an observed outcome Y, the complete-label coefficient vector is

$$\boldsymbol{\beta}(y) = B(y)^{-1} m(y), \qquad m_r(y) = y^{\mathsf{T}} C^{(r)\mathsf{T}} M_V Y, \qquad B_{rs}(y) = y^{\mathsf{T}} C^{(r)\mathsf{T}} M_V C^{(s)} y, \tag{C6}$$

when $B(y)$ is nonsingular. The same item label enters every component of m and B, so their audit uncertainty is joint. If Y is also built from these labels, its numerator moments are quadratic too. This is the usual regression normal equation with its item-level construction made explicit; it does not prescribe one particular correction method.

The fixed-sample coefficient in equation (3) has a linear numerator and a quadratic denominator in the item labels. Put $r = y - q$. Then

$$\begin{aligned} N &= y^{\mathsf{T}} a = q^{\mathsf{T}} a + r^{\mathsf{T}} a, \\ D &= y^{\mathsf{T}} G y = q^{\mathsf{T}} G q + 2 r^{\mathsf{T}} G q + r^{\mathsf{T}} G r. \end{aligned} \tag{C7}$$

Let R_j indicate an item's inclusion in a probability audit, with $\pi_j = \Pr_R(R_j = 1) > 0$ and $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1) > 0$ for every relevant pair. Set $\pi_{jj} = \pi_j$. The following estimates are design unbiased for the two moments, conditional on C, V, Y, y, q:

$$\begin{aligned} \widehat{N} &= q^{\mathsf{T}} a + \sum_j \frac{R_j}{\pi_j} r_j a_j, \\ \widehat{D} &= q^{\mathsf{T}} G q + 2 \sum_j \frac{R_j}{\pi_j} r_j (Gq)_j + \sum_{j,\ell} \frac{R_j R_\ell}{\pi_{j\ell}} r_j G_{j\ell} r_\ell. \end{aligned} \tag{C8}$$

The identities follow from $\mathrm{E}_R[R_j/\pi_j] = 1$ and $\mathrm{E}_R[R_j R_\ell/\pi_{j\ell}] = 1$. No independent-item-error or calibrated-score assumption is used. The ratio $\widehat{N}/\widehat{D}$ is generally not design unbiased and can be unstable or undefined when the estimated denominator is near zero or nonpositive. Valid inference for the ratio requires additional design analysis; its variance involves inclusion probabilities for sets of up to four items. The formulas identify the information an item audit would have to preserve. The linear-label difference estimator in equation (B1) does not by itself correct a regression that uses Cy as a covariate.

For a retrospective standard error, treat $(\widehat{N}, \widehat{D})$ as a joint vector of item- and item-pair Horvitz–Thompson totals. If the required inclusion probabilities for sets of up to four items are known and positive, its design covariance can be estimated from the realized audit. When D is bounded away from zero and a normal approximation for the joint moments is justified, the delta-method variance

for N/D uses gradient $(1/D, -N/D^2)^\mathsf{T}$. Near a zero denominator, a ratio confidence set must instead respect the joint uncertainty in numerator and denominator; the plug-in delta standard error is unreliable. With several constructed regressors, the corresponding moment vector includes all residualized regressor–outcome and regressor–regressor cross-products, including equation (C5). Its covariance and the regression map must be propagated jointly. These are routes to inference under an observed probability-validation design, not a claim that an arbitrary validation set supports an automatic corrected standard error.

D Repeated-prediction error and dependence

For a fixed fitted classifier and fixed labels, Δ is fixed. To discuss bias and variance, specify a repeated-prediction process, such as repeated training samples, while fixing y, w and the inputs that determine w. All moments below condition on this fixed design. Define $b = \mathrm{E}[e]$ and the full joint covariance $\Sigma = \mathrm{Cov}(e)$, without assuming independence across items. Then

$$\begin{aligned}
\mathrm{Bias}(\widehat{\theta}) &= w^{\mathsf{T}} b, \\
\mathrm{Var}(\Delta) &= w^{\mathsf{T}} \Sigma w = \sum_j w_j^2 \Sigma_{jj} + 2 \sum_{j<\ell} w_j w_\ell \Sigma_{j\ell}, \qquad \text{(D1)} \\
\mathrm{E}[\Delta^2] &= (w^{\mathsf{T}} b)^2 + w^{\mathsf{T}} \Sigma w. \qquad \text{(D2)}
\end{aligned}$$

Equations (1)–(D2) connect the research quantity to its item weights, systematic error, and uncertainty. Covariance between item errors enters the estimate's variance. Dependence alone does not determine bias; systematic errors on items with nonzero weights can make $w^{\mathsf{T}} b$ nonzero.

Unit aggregates and regression vectors are corresponding joint quantities. Writing $\widehat{T} = Cq$, $d = \widehat{T} - T$, and $\widehat{\beta} = ACq$ gives

$$\begin{aligned}
d &= Ce, \qquad \text{(D3)} \\
\mathrm{E}[d] &= Cb, \qquad \mathrm{Cov}(d) = C\Sigma C^{\mathsf{T}}, \qquad \text{(D4)} \\
\mathrm{E}[\widehat{\beta} - \beta^*] &= ACb, \qquad \mathrm{Cov}(\widehat{\beta} - \beta^*) = AC\Sigma C^{\mathsf{T}} A^{\mathsf{T}}. \qquad \text{(D5)}
\end{aligned}$$

The same item error is reused across every unit linked to that item. These formulas quantify the specified prediction-error process; they do not include errors in the item-to-unit map, unit sampling, or uncertainty about the construct. They concern a label-derived aggregate as an outcome. Using that aggregate as a regressor is a different errors-in-variables problem treated in Section 2.

D.1 Shared labels and correlated item errors

These are separate mechanisms under a repeated-prediction model. First, a single item error can appear in several units' aggregates. Even if the item errors e_j are independent across j, units that share an uncertain item can have correlated aggregate errors. This is a standard common-shock structure, analogous to shift-share inference (Adão et al., 2019); it does not assert correlation between distinct item errors. Second, distinct item errors may themselves covary under repeated fitting. Shared training data or model parameters can produce such covariance, but using one fitted classifier does not by itself establish it. Units linked to disjoint items can have correlated aggregate errors only under this second mechanism or another shared source of uncertainty. In equation (D1), positively correlated item errors increase variance when their weights have the same sign, and decrease it when their weights have opposite signs.

For a limiting example, suppose centered errors are shared within item families: $e_j - b_j = u_{g(j)}$, with independent family shocks u_g of variance τ_g^2. This is compatible with binary predictions when the classifier assigns one common random prediction to every item in a family. Then

$$\mathrm{Var}(\Delta) = \sum_g \tau_g^2 \left(\sum_{j:g(j)=g} w_j \right)^2. \qquad \text{(D6)}$$

A family of m items with unit weights contributes $m^2 \tau_g^2$, whereas assuming independent item errors would give $m\tau_g^2$. Signed weights can instead cancel within a family. The relevant concentration is in

aggregate family influence, not just individual item influence. This example is a special covariance structure, not an assertion that the right families are known or that all within-family mistakes are perfectly correlated.

Independent item errors are another special case: Σ is diagonal, with entries v_j. For any unit aggregate $T = Cy$, equation (D4) reduces to

$$\mathrm{Var}(d_i) = \sum_j c_{ij}^2 v_j, \qquad \mathrm{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} v_j. \tag{D7}$$

Let α_j be the false-positive probability on a truly negative item and β_j the false-negative probability on a truly positive item. For binary predictions, $v_j = \alpha_j(1-\alpha_j)$ on negative items and $v_j = \beta_j(1-\beta_j)$ on positive items. In browsing, repeated visits multiply one classification error; treating visits as independent would incorrectly replace squared visit weights by visit counts. More generally, any reuse of an uncertain item can induce dependence across units.

The same predictions can build several different aggregates. If the rth aggregate uses a fixed matrix $C^{(r)}$, then

$$d^{(r)} = C^{(r)} e, \qquad \mathrm{Cov}\{d^{(r)}, d^{(s)}\} = C^{(r)} \Sigma C^{(s)\mathsf{T}}. \tag{D8}$$

For scalar estimates with item-weight vectors $w^{(r)}$, independent item errors yield $\mathrm{Cov}(\Delta_r, \Delta_s) = \sum_j v_j w_j^{(r)} w_j^{(s)}$. Thus each item contributes a rank-one term to the joint covariance of all estimates. A heavily weighted item can move several reported quantities together; subtracting two of them can cancel its error. For a sequence of larger unit samples, independent item errors with $v_j\{w_j^{(r)}\}^2$ bounded away from zero for even one item prevent the prediction-error variance of that estimate from vanishing. More units alone do not supply more independent predictions for that item. The covariance statements require a repeated-prediction model; $d^{(r)} = C^{(r)} e$ remains an exact identity for fixed realized predictions.

For a fixed linear unit contrast $a^\mathsf{T} d$, equal item-error variances give a simple overlap diagnostic:

$$R(a, C) = \frac{\sum_j (\sum_i a_i c_{ij})^2}{\sum_i a_i^2 \sum_j c_{ij}^2}. \tag{D9}$$

The numerator is its variance, apart from the common variance factor. The denominator is what that variance would be if each unit's aggregate error were independent while retaining its marginal variance. Thus R measures the effect of discarding covariance across units for this one prediction-error component. It is at least one for nonnegative weights and counts, but may be above or below one for a signed contrast. It is not a general standard-error correction: unequal item variances, correlated item errors, unit sampling, and other errors require their own treatment.

An uncertainty procedure must follow the same structure as the estimand. Resampling downstream records while holding predictions fixed does not propagate repeated-training uncertainty. Each replicate must preserve reused labels across records and the joint distribution of errors across items. Independently perturbing each item label imposes a diagonal Σ; resampling the classifier fit and predicting all items together can preserve dependence induced by that fitting process, provided the resampling scheme represents that process. A prediction-error covariance cannot simply be added to an arbitrary reported regression covariance: the components may overlap or be dependent. Neither larger downstream samples nor clustering records alone guarantees that this error disappears, and standard errors do not repair bias.

D.2 Many items do not ensure a normal approximation

Even with independent item errors, diffuse variance shares alone do not justify Gaussian inference. Let all true labels be zero and let each of k items be falsely labeled positive independently with

probability $1/k$. With $w_j = 1$, total error B_k is binomial with mean one and variance $1 - 1/k$. The largest item variance share is $1/k$, yet $B_k - 1 \Rightarrow \text{Poisson}(1) - 1$, not a normal distribution. At $k = 10{,}000$, an oracle interval using the exact bias and variance but a normal 1.96 cutoff has coverage 91.97%. This calculation uses exact binomial probabilities, not simulation.

For independent bounded binary errors, a sufficient Lindeberg condition is

$$\frac{\max_j |w_j|}{\sqrt{\sum_j w_j^2 v_j}} \to 0.$$

A small maximum variance share $\max_j w_j^2 v_j / \sum_j w_j^2 v_j$ is weaker when errors become rare. The example violates the former condition and satisfies the latter. A growing number of items cannot substitute for a condition on the distribution of their errors. This is a prediction-error example, distinct from the fixed-label audit design.

E Count-certificate construction and audit diagnostics

Partition items into strata $h = 1, \dots, H$ before observing their audited labels. Sample n_h of the M_h items without replacement. If K_h is the unknown number of incorrect predictions and X_h the sampled error count, then $X_h \sim \text{Hypergeometric}(M_h, K_h, n_h)$. Assign nonnegative error budgets δ_h with $\sum_h \delta_h \le \delta$. Retain integers K for which both $\Pr_K(X_h \le x_h) \ge \delta_h/2$ and $\Pr_K(X_h \ge x_h) \ge \delta_h/2$. Their extrema are ℓ_h, u_h. A census stratum has $\ell_h = u_h = x_h$; an unaudited stratum contributes no count restriction.

Let O be the audited items, U_h the unaudited items in stratum h, $c_j = (2q_j - 1)w_j$, and $d_O = \sum_{j \in O} w_j(q_j - y_j)$. The smallest and largest compatible errors are

$$L_A = d_O + \sum_h \min_{\ell_h - x_h \le m \le u_h - x_h} S_-(c_{U_h}, m),$$
$$U_A = d_O + \sum_h \max_{\ell_h - x_h \le m \le u_h - x_h} S_+(c_{U_h}, m). \tag{E1}$$

The optimization is over integers. With signed weights, an extreme need not occur at the largest admissible error count.

Proof of Proposition 3. Inverting each pair of hypergeometric tails excludes the true K_h with probability at most δ_h. A union bound therefore includes all true counts with probability at least $1-\delta$. On that event, the true errors in U_h number between $\ell_h - x_h$ and $u_h - x_h$. Sorting and choosing a count in that range gives the extrema in equation (E1); the choices are independent restrictions across strata and hence jointly attainable. The true label vector lies in the same configuration set for every w, so the coverage event applies simultaneously to all its linear contrasts. □

The labels and predictions are fixed under this design. Systematic or correlated classifier mistakes do not invalidate coverage, which is over repeated audit samples. The count region can be conservative, and sharp projection of that region does not make its interval shortest. The guarantee concerns one label vector; combining separate audits of several outcomes needs an additional error allocation. Optional stopping, adaptive strata, and imperfect audit labels require other procedures or assumptions.

Information lost by count projection. Let all N predictions equal one, half the weights equal $1/N$ and half equal $-1/N$. Suppose a fraction $p < 1/2$ of labels in each half are wrong. The true contrast is zero. If $n \to \infty$ but $n/N \to 0$, an unstratified audit consistently learns the overall error fraction. Yet unaudited errors can still be placed almost entirely on either weight sign: the projected interval width converges to $2p$. A direct estimate of the weighted audit residual instead has standard error of order $n^{-1/2}$. Separating the two weight signs into strata removes this particular information loss.

For a declared contrast, the stratified difference estimator uses $z_j = w_j(y_j - q_j)$ directly:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j \in O_h} z_j, \qquad \widehat{V} = \sum_h \frac{M_h^2(1 - n_h/M_h)}{n_h} s_{z,h}^2. \tag{E2}$$

Under independent stratum draws, the estimate and variance estimator are design unbiased when each noncensus stratum has at least two sampled items. Normal intervals need a separate approximation. A conservative finite-sample alternative applies Hoeffding's without-replacement inequality (Hoeffding, 1963) to each bounded residual mean and sums the stratum radii. More efficient weighted audit procedures exist (Shekhar et al., 2023).

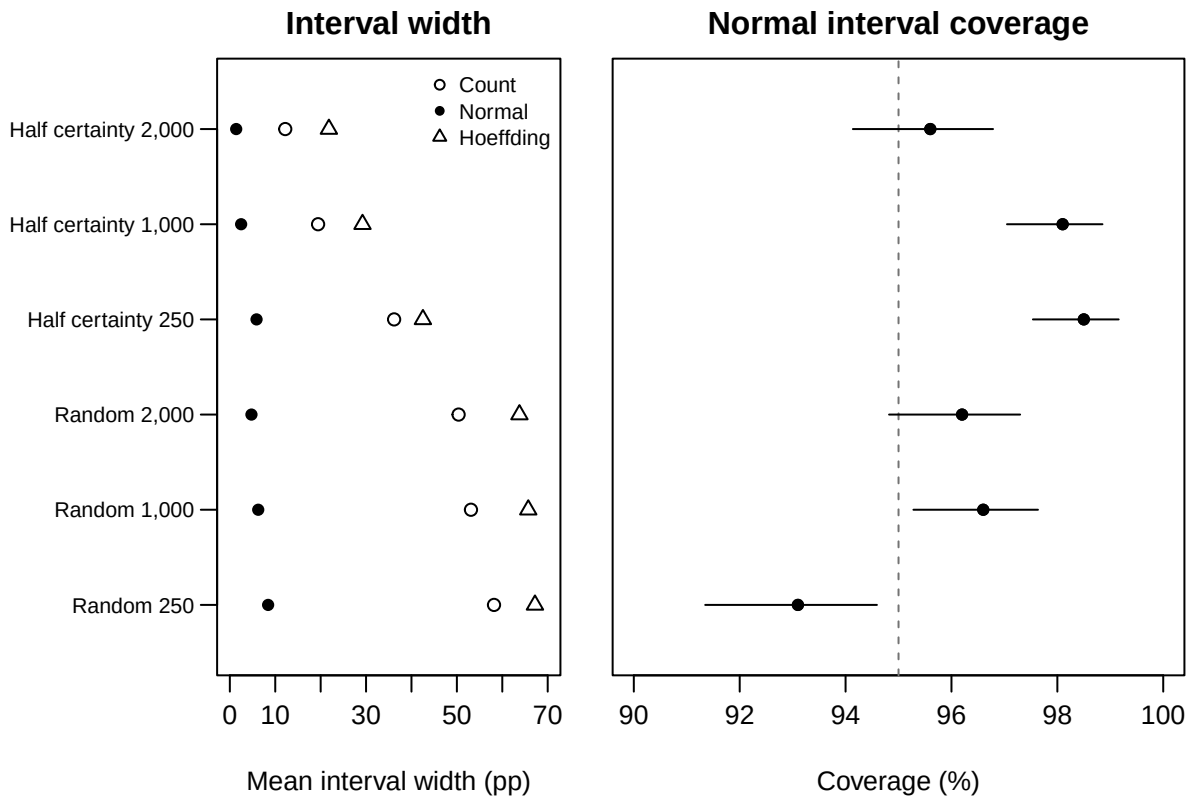


Figure E1: Width and coverage for the Facebook-pixel female share coefficient. Rows compare audit budgets and designs in the fixed browsing frame. Left: mean interval width in percentage points over 1,000 audits. Right: normal interval coverage, with exact 95% binomial Monte Carlo intervals and a dashed 95% reference. Count and Hoeffding intervals covered in every simulated draw. "Half certainty" audits the largest absolute influences with half the budget. Count intervals cover all contrasts of this outcome simultaneously; direct intervals target this coefficient. The randomness is domain auditing; users and predictions are fixed.

Table E1: All audit designs and methods for the focal coefficient

Design	Budget	Interval	Coverage (%)	Width (pp; MCSE)	Correct sign (%)	Wrong sign (%)
Random	250	Count projection	100.0	58.2 (0.06)	0.0	0.0
Random	250	Direct normal	93.1	8.4 (0.50)	39.6	0.0
Random	250	Direct Hoeffding	100.0	67.2 (0.01)	0.0	0.0
Random	1,000	Count projection	100.0	53.1 (0.05)	0.0	0.0
Random	1,000	Direct normal	96.6	6.3 (0.19)	32.9	0.0
Random	1,000	Direct Hoeffding	100.0	65.7 (0.02)	0.0	0.0
Random	2,000	Count projection	100.0	50.4 (0.05)	0.0	0.0
Random	2,000	Direct normal	96.2	4.8 (0.11)	34.4	0.0
Random	2,000	Direct Hoeffding	100.0	63.8 (0.03)	0.0	0.0
Half certainty	250	Count projection	100.0	36.2 (0.05)	0.0	0.0
Half certainty	250	Direct normal	98.5	5.9 (0.29)	40.6	0.0
Half certainty	250	Direct Hoeffding	100.0	42.5 (0.00)	0.0	0.0
Half certainty	1,000	Count projection	100.0	19.4 (0.04)	0.0	0.0
Half certainty	1,000	Direct normal	98.1	2.5 (0.04)	41.9	0.0
Half certainty	1,000	Direct Hoeffding	100.0	29.2 (0.00)	0.0	0.0
Half certainty	2,000	Count projection	100.0	12.2 (0.03)	0.0	0.0
Half certainty	2,000	Direct normal	95.6	1.4 (0.01)	65.0	0.0
Half certainty	2,000	Direct Hoeffding	100.0	21.8 (0.00)	0.0	0.0

Each row summarizes 1,000 domain audits. Width is the mean in percentage points, with its Monte Carlo standard error in parentheses. Sign columns report exclusion of zero in the correct or wrong direction. All methods within a design use the same draws. Exact binomial Monte Carlo intervals and replicate results accompany the package.

The random and half-certainty designs were chosen for one coefficient. They do not establish that this design is efficient for other contrasts. The count certificate covered the target in every simulated draw but never excluded zero. Direct normal intervals were much narrower, yet coverage in the smallest random design fell below the nominal level. The finite-sample Hoeffding baseline was wider than count projection. These comparisons do not identify a finite-sample-valid, efficient direct interval for this design.

Table E2: Refining validation strata for the focal coefficient

Strata	Exact-count width (pp)	Audit width (pp; MCSE)	Coverage (%)
2	49.7	53.2 (0.05)	100.0
10	34.7	47.1 (0.05)	100.0
50	21.8	48.2 (0.03)	100.0

Rank bins of signed influence cross predicted labels. Exact-count width conditions on the population count in every stratum; audit width averages 95% count-projection intervals over 1,000 domain samples of size 1,000. Width uncertainty is its Monte Carlo standard error. These partitions were examined after the initial audit results. No design certified the sign.

The narrower exact-count range from finer strata need not translate into a narrower audit interval: the fixed sample is divided among more strata and the error budget is split across their discrete hypergeometric intervals. The 10- versus 50-stratum mean-width difference should be read with the reported Monte Carlo uncertainty, not as evidence of an optimal partition.

F Retaining the highest-visit domain labels

Table F1 repeats the count-information exercise after retaining both observed classification labels for the highest-visit domains. For the remaining domains it retains only the numbers positive under Piedomains alone and under the provider alone. The ordering uses visit count with domain name to break ties. The count range is sharp under exactly this information: the head contribution is fixed, while the tail discrepancies can occupy any compatible tail domains. At zero retained domains it equals the range in Table 2. The provider's observed visit share lies inside every range.

Table F1: Count-compatible provider visit shares after retaining label pairs for the highest-visit domains

Provider	Domains retained	Visits retained (%)	Count range (%)	Width (pp)
Bitdefender	0	0.0	[0.07, 82.11]	82.05
Bitdefender	10	42.2	[0.07, 40.06]	39.99
Bitdefender	50	56.8	[0.07, 25.80]	25.73
Bitdefender	250	71.5	[1.00, 13.65]	12.64
Bitdefender	1,000	83.4	[1.33, 5.98]	4.64
Forcepoint	0	0.0	[0.07, 80.61]	80.54
Forcepoint	10	40.6	[0.07, 40.16]	40.09
Forcepoint	50	55.1	[0.07, 25.99]	25.92
Forcepoint	250	69.6	[0.86, 14.01]	13.15
Forcepoint	1,000	81.9	[1.24, 6.24]	4.99
alphaMountain	0	0.0	[0.04, 68.07]	68.03
alphaMountain	10	30.1	[0.04, 38.33]	38.30
alphaMountain	50	49.0	[0.56, 21.37]	20.82
alphaMountain	250	67.2	[1.44, 9.12]	7.68
alphaMountain	1,000	81.5	[1.87, 4.03]	2.16

The percentages use each provider row's categorized-domain visit total as its denominator. "Visits retained" is the share of that total on domains whose two observed labels remain known. Width is in percentage points. The providers use different category definitions and neither label is an adjudicated content truth. Retaining a label pair here is an information exercise, not a claim that a manual audit was performed.